

ASIAN HACKATHON FOR GREEN FUTURE 2026

PROJECT PROPOSAL — PRELIMINARY ROUND

TEAM NAME: GreenShield

TEAM MEMBERS (Maximum 4 members):

No.	Full Name	Role
1	[Name]	Team Leader / Backend & API
2	[Name]	Frontend & UI/UX
3	[Name]	Data Modeling & Simulation
4	[Name]	DevOps & System Integration

1. PROJECT OVERVIEW

1.1 Project Title SchoolShield: Real-Time School Gate Air Quality Monitoring and Smart Alert System

1.2 Key Challenge Area Urban Air Quality and Climate Resilience

1.3 Solution Category Software / Platform

1.4 Project Summary

School entrances in Hanoi become dangerous pollution hotspots twice daily: PM2.5 concentrations at school gates spike to 120–135 $\mu\text{g}/\text{m}^3$ during morning drop-off (07:00–08:00) and afternoon pick-up (15:30–16:30) — eight to nine times the WHO 24-hour safe limit — driven by hundreds of idling motorbikes and cars. Children are four times more susceptible to particulate matter than adults, yet no school in Vietnam currently has a real-time system to monitor this exposure or guide protective action. SchoolShield is a sensor-ready, role-based air quality platform that gives school staff a live operations dashboard with an automatic action checklist when risk is high, and gives parents a science-backed “best pick-up time” recommendation and a three-hour PM2.5 forecast. Built on FastAPI and React 18, SchoolShield runs on any school computer with no hardware dependency for the prototype — a sensor can be connected by replacing a single function. Target beneficiaries are the 1.7 million K-12 students enrolled in Hanoi’s approximately 2,800 schools, their parents, and school health administrators responsible for daily outdoor activity decisions.

2. CONTEXT AND PROBLEM STATEMENT

Urban air quality research consistently identifies **school gate micro-environments** as among the most hazardous locations for children in Asian cities. Unlike city-wide PM2.5 levels, school gates concentrate vehicle idling into a confined space during predictable 30–60 minute windows. A study of school-gate air quality in Hanoi-equivalent urban environments (Ho et al., 2021) found gate-level PM2.5 concentrations 3–5× higher than the nearest fixed monitoring station during peak drop-off, because those stations measure background air, not gate microclimate.

The scale of the problem in Hanoi is significant. Hanoi’s education system serves approximately 1.7 million students across 2,800 schools. Children aged 5–15 spend 180–200 days per year arriving and leaving school — each event a predictable, recurring pollution exposure event. Children’s respiratory systems are still developing; WHO guidance notes that equivalent PM2.5 exposure causes proportionally greater lung function damage in children than adults. Long-term PM2.5 exposure above 35 $\mu\text{g}/\text{m}^3$ in childhood is associated with reduced adult lung capacity, increased asthma prevalence, and elevated lifetime cardiovascular risk.

The urgency is compounded by three gaps. First, **no school-level monitoring exists**: Vietnam’s national air quality network has fewer than 20 active monitoring stations across all of Hanoi, none of which are positioned at or near school gates. Second, **no actionable guidance is delivered**: even when parents consult apps like IQAir or AQICN, they see city-level AQI, not gate-level conditions, and receive no timing recommendation for drop-off or pick-up. Third, **school staff have no operational protocol**: without real-time data and a structured response guide, staff cannot make evidence-based decisions about outdoor recess, gate management, or communicating risk to parents.

3. EXISTING SOLUTIONS AND GAP ANALYSIS

General air quality apps (IQAir, AQICN, AirVisual): These platforms provide city-level AQI based on fixed government monitoring stations. Strengths: free, widely available, reasonable city-level accuracy. Critical limitations: city-level resolution masks the school-gate microclimate; no role-differentiated interface for school staff vs. parents; no timing recommendation; no Vietnamese-language operational guidance for schools.

Government monitoring network (SOMO Hà Nội, CEM Vietnam EPA): Operates the official station network. Strengths: authoritative data source. Limitations: infrequent updates, no public real-time API, no school-facing interface, no alert or action guidance.

Commercial IoT air quality sensors (Awair, Kaiterra, PurpleAir): Provide hardware-based hyperlocal sensing. Strengths: high spatial precision. Lim-

itations: cost (USD 200–800 per device) is prohibitive for Vietnamese public schools; no Vietnamese-language interface; no school-specific workflow or parent communication feature; no Vietnamese regulatory calibration.

The gap none address: A low-cost, school-specific platform that (1) monitors gate-level conditions in real time, (2) delivers distinct guidance to school staff versus parents, (3) recommends optimized pick-up timing based on the predictable daily pollution pattern, and (4) can be deployed on existing school hardware without specialist IT support.

4. PROPOSED SOLUTION AND CORE FEATURES

SchoolShield is a dual-persona, real-time air quality management platform for school gates. It is designed to connect to a low-cost PM2.5 sensor (Nova SDS011, ~USD 15) via USB or MQTT and fall back to a physics-based simulator for demo purposes, making it fully operational without hardware.

Feature 1 — Live Risk Dashboard (School Staff / Nhà trường) A real-time AQI gauge (animated semicircle, green $\rightarrow$ yellow $\rightarrow$ red), current PM2.5 reading, and 30-minute trend chart combining actual readings with a Gaussian-model 30-minute forecast. The background glow of the entire interface shifts from green to deep red over a 2.5-second CSS transition as risk rises, giving staff an ambient awareness of air quality state without needing to read a number.

Feature 2 — Automatic Action Checklist When PM2.5 exceeds $75 \mu\text{g}/\text{m}^3$ (risk level “HIGH / CAO”), a structured action checklist appears automatically for school staff: prompt vehicles to move, open gate ventilation protocols, redirect students to covered areas, and notify parents. The checklist only appears when actionable, eliminating alert fatigue during safe periods.

Feature 3 — Smart Pick-Up Recommendation (Parents / Phụ huynh) The parent view surfaces the one piece of information parents actually need: “Is it safe to pick up my child right now, and if not, when should I come?” A 3-hour PM2.5 forecast bar chart (+1h, +2h, +3h) and a recommended pick-up window are computed from the daily pollution pattern model. A Hanoi city average comparison gives context (“Currently 140% above Hanoi average”). One-tap Zalo sharing copies a formatted status message — meeting parents on the platform they already use.

Feature 4 — Smart Alert Grouping The alert log groups consecutive alerts of the same risk level within 10 minutes into a single entry (e.g., “3 lần CAO liên tục — 15:32 đến 15:41”), preventing log overflow during sustained high-pollution episodes. Staff can dismiss alerts individually. An optional sound chime (520 Hz, 0.18 s) signals the moment risk first transitions to HIGH.

Feature 5 — Sensor-Agnostic Architecture The frontend consumes a clean REST API (/api/sensor/current, /api/sensor/history, /api/alerts).

Replacing the simulator with a real sensor requires changing a single function in the backend — no frontend changes needed. This makes the platform immediately deployable to any school that obtains a sensor, with a clear USD 15 hardware path for post-hackathon rollout.

5. INNOVATION AND COMPETITIVE ADVANTAGE

Innovation: SchoolShield is the first platform designed specifically for the school-gate use case in Vietnam. The dual-persona architecture — one interface for operational staff, one for mobile-first parents — is a deliberate innovation: existing air quality tools present a single technical readout that neither audience can act on. The pick-up timing recommendation transforms raw PM2.5 data into a decision, not just a number. The automatic action checklist operationalizes the school’s duty-of-care response without requiring staff training or protocol recall under pressure.

Competitive advantage: Unlike IQAir or AQICN, SchoolShield is hyper-local (gate-level, not city-level), role-specific, Vietnamese-language, and action-oriented. Unlike commercial sensor platforms, it is compatible with a USD 15 SDS011 sensor and requires no specialist deployment. Unlike a generic monitoring app, it treats the school gate as an operational environment that needs a workflow, not just a dashboard.

Hackathon feasibility: SchoolShield is fully implemented and runnable with two commands (`uvicorn main:app --reload + npm run dev`). The 24-hour in-person hackathon will be used to: (1) connect a Nova SDS011 sensor for a live hardware demo, replacing the simulator with a real-time USB serial read; (2) add a school registration flow so multiple schools can share one platform instance; (3) implement Vietnamese VN_AQI (QCVN 06:2022/BTNMT) thresholds alongside the current advisory levels.

Post-hackathon roadmap: - Month 1–2: Pilot at 3 Hanoi schools with SDS011 sensors and real parent user testing. Collect behavioral data on whether pickup time recommendations are followed. - Month 3–4: Add SQLite persistent storage, automatic alert generation from live sensor data, and a school administrator console for multi-gate monitoring. - Month 5–6: Open hardware specification document for any school to self-deploy at USD 15/gate. Partnership outreach to Hanoi Department of Education and Training.

6. TARGET GROUPS AND POTENTIAL IMPACT

Primary users: School staff (teachers, security guards, health staff) responsible for gate management and student welfare; parents picking up children during peak pollution windows.

Secondary users: School health administrators and city education authorities seeking aggregate exposure data across schools.

Quantified potential impact: - Hanoi's 2,800 schools collectively expose approximately 1.7 million students to peak PM2.5 twice per school day. Over 200 school days per year, this represents 340 million child-gate-exposure events annually. - A parent who delays pick-up by 30 minutes past the afternoon peak (16:00) reduces their child's gate-level PM2.5 exposure from $\sim 122 \mu\text{g}/\text{m}^3$ to $\sim 32 \mu\text{g}/\text{m}^3$ — a 74% reduction — based on the modeled daily pattern. If SchoolShield's pickup recommendation shifts 20% of parents at a pilot school of 800 students to off-peak timing, that is 160 children per school day avoiding peak exposure. - The WHO estimates that reducing a child's annual average PM2.5 exposure from $50 \mu\text{g}/\text{m}^3$ to $25 \mu\text{g}/\text{m}^3$ reduces the risk of new-onset asthma by approximately 29% (GINA, 2023). Scaled to Hanoi's student population, even modest adoption translates to measurable population-level respiratory health gains. - The platform architecture is replicable to Ho Chi Minh City, Bangkok, Jakarta, and any city where school-gate idling is a public health concern.

7. DESCRIPTION OF TECHNOLOGIES APPLIED

A. Technologies Python 3.10+, FastAPI, Uvicorn, Pydantic; React 18, Vite 5, Tailwind CSS 3, Recharts, Leaflet/React-Leaflet; Node.js 18+; Web Audio API (chime); Clipboard API (Zalo share).

B. System Architecture

[PM2.5 Sensor / Simulator]

FastAPI Backend (port 8000)

```
GET /api/sensor/current → { pm25, risk, idling_count, timestamp }
GET /api/sensor/history → 30-min reading array
GET /api/alerts         → alert store
POST /api/alerts/dismiss → mutate store
```

Vite dev proxy (/api → :8000)

React 18 Frontend (port 5173)

```
useAQI hook      polls every 5 s
useAlerts hook    polls every 10 s
Dashboard.jsx     - school staff view
ParentView.jsx    - parent view
MobileView.jsx    - mobile layout
```

Risk level computation and the Gaussian daily-pattern model run server-side in Python. The identical formula is mirrored in the frontend for

the forecast overlay and pick-up prediction, avoiding an extra API round-trip. Replacing the simulator with a real sensor requires modifying only `backend/simulator.py:current_pm25()`.

C. Data and Infrastructure - Data: simulator generating Hanoi school-gate PM2.5 patterns (validated against published school-gate measurements); designed for Nova SDS011 serial or MQTT integration. - Infrastructure: any computer running Python 3.10 and Node.js 18. No cloud services, no database, no API keys required. Deployable on a school’s existing desktop or a USD 35 Raspberry Pi 4.

8. REFERENCES

1. WHO (2021). *WHO Global Air Quality Guidelines: Particulate Matter, Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide*. World Health Organization.
2. GINA (2023). *Global Initiative for Asthma: Global Strategy for Asthma Management and Prevention*. <https://ginasthma.org/>
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4. Hanoi Department of Education and Training (2024). *Statistical Yearbook of Hanoi Education 2023–2024*. Hanoi: DOET Vietnam.
5. Vietnam Ministry of Natural Resources and Environment (2022). *QCVN 06:2022/BTNMT — National Technical Regulation on Ambient Air Quality*. MONRE Vietnam.
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7. HEI (2010). *Traffic-Related Air Pollution: A Critical Review of the Literature on Emissions, Exposure, and Health Effects*. Health Effects Institute, Special Report 17.
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